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Syed Talha Bukhari

Research Interests

Machine Learning, Computer Vision, Robotics

Education

2018 *to* 2020 **M.S. Electrical Engineering**, *CGPA: 3.96/4.00*,
LUMS School of Science and Engineering (LUMS SSE), Lahore.

- Graduated in Dean's Honor List.
- Thesis/Research Focus: Deep Learning for 3D mpMRI Segmentation.
- Coursework includes: Convex Optimization, Stochastic Systems, Information Theory & Machine Learning, Computer Vision Fundamentals, Deep Learning, Advanced Digital Signal Processing, Robot Motion Planning.

2014 *to* 2018 **B.S. Mechatronics & Control Engineering**, *CGPA: 3.94/4.00*,
University of Engineering and Technology (UET), Lahore.

- Graduated with highest distinction, UET Gold Medal awardee, Best Senior Year Project (thesis).
- Coursework includes: Robotics, Control Systems, Digital Image Processing, Digital Signal Processing, Signals & Systems, Intelligent Systems, Embedded Systems, Numerical Methods, Hydraulics & Pneumatics.

Work Experience

8/2021 *to* **Present** **Machine Learning Software Engineer**, Pointivo, Inc., USA.

- Working, as part of the Cognitive Services team, to deploy Deep Learning-based approaches to automate physical asset inspection via Machine Vision.

7/2020 *to* 9/2021 **Research Associate**, *Department of Electrical Engineering*,
LUMS School of Science and Engineering, Lahore.

- Led the development of Deep Learning pipelines for medical image segmentation.
- Co-supervised and guided M.S. and Ph.D. students working in the group.
- Lab Director/Supervisor: Dr. Hassan Mohy-ud-Din

1/2019 *to* 6/2020 **Graduate Research Assistant**, *Department of Electrical Engineering*,
LUMS School of Science and Engineering, Lahore.

- Led the development of Deep Learning pipelines for medical image segmentation.
- Part-time employment, in parallel with graduate studies.
- Lab Director/Supervisor: Dr. Hassan Mohy-ud-Din

Spring 2020 **Graduate Teaching Assistant**, *EE516: Foundations of Deep Learning*,
LUMS School of Science and Engineering, Lahore.

- Designed coursework and graded components with the instructor, in the inaugural offering of the graduate course.
- Focused on imparting fundamental skills relevant to implementation of Deep Learning in PyTorch.

07 *to* 08/2017 **Intern**, *Centre of Excellence in Solid State Physics*,
University of the Punjab, Lahore.

- Involved in device fabrication and characterization process at micro level.
- Supervisor: Prof. Dr. Shahzad Naseem

Technical Skills

Languages Python, MATLAB, C++, Kotlin, Julia

Frameworks PyTorch, TensorFlow (1.x & 2.x), pandas, OpenCV, JAX (basics)

Embedded Proteus (ISIS & ARES), Arduino, MPLab (basics)

Robotics V-REP (with MATLAB), ROS (basics)

CAD SolidWorks, AutoCAD

Other LaTeX, Git, Docker, Bash scripting

Talks/Workshops

11/2021 **Participant**, *Online Asian Machine Learning School (OA^{M}LS)*, The 13th Asian Conference on Machine Learning (ACML 2021).

- Selected from a pool of applicants from the Asia-Pacific region.
- A two-week interactive session, providing exposure to topics in theoretical and applied machine learning.

11/2020 **Speaker**, *Seminar on Deep Learning*, University of Engineering & Technology, Lahore.

- A one-day seminar covering fundamental concepts in Machine Learning and Deep Learning, targeted at senior undergraduate students.

12/2019 **Co-Instructor**, *Short Course on Machine Learning and Deep Learning*, 2nd Annual Workshop on Big Data and Cloud Computing (BigC 2019).

- A one-day workshop/seminar covering fundamental concepts in Machine Learning and Deep Learning, and implementation of Neural Networks in Python with TensorFlow.

Awards & Honors

2020 LUMS Dean's Honor Roll for distinction in graduate studies.

2018–20 LUMS Merit Scholarship Award for graduate studies (100% tuition waiver).

2018 UET Gold Medal for top rank in undergraduate studies.

2014–18 UET Dean's Honor Roll (×8) for distinction in undergraduate studies.

2018 1st Position Senior Year Project (awarded by department): Automatic Retail Checkout via Deep Learning.

2017–18 Senior Year Project approved for funding by National Grassroots ICT Research Initiative.

2014 College Best Student of the Year (12th grade).

2008 International Kangaroo Mathematics Contest: 27th position in the country (3rd in school) out of 10,556 participants at the *Benjamins* level.

Publications

Jabbar, Sana ; Bukhari, Syed T. ; Mohy-ud-Din, Hassan: On the Generalizability of Convolutional Neural Networks for Accurate Quantification of Cardiac Function on 3D MRI Scans: A Multi-vendor, Multi-center Study. In: *Physics in Medicine & Biology* (**under review**)

Bukhari, Syed T. ; Din, Hassan M.: E_1D_3 U-Net for Brain Tumor Segmentation: Submission to the RSNA-ASNR-MICCAI BraTS 2021 Challenge. In: *International MICCAI Brainlesion Workshop* Springer, 2021. – (Available at https://arxiv.org/abs/2110.02519)

Jabbar, Sana ; Bukhari, Syed T. ; Mohy-ud-Din, Hassan: Multi-view SA-LA Net: A Framework for Simultaneous Segmentation of Right Ventricle on Multi-View Cardiac MR Images. In: *International Workshop on Statistical Atlases and Computational Models of the Heart (STACOM 2021)* Springer, 2021. – (Available at https://arxiv.org/abs/2110.00682)

Bukhari, Syed T. ; Mohy-ud-Din, Hassan: A Systematic Evaluation of Learning Rate Policies in Training CNNs for Brain Tumor Segmentation. In: *Physics in Medicine & Biology* [IF:3.61] (2021). https://iopscience.iop.org/article/10.1088/1361-6560/abe3d3

Bukhari, Syed T. ; Amin, Abdul W. ; Naveed, Muhammad A. ; Abbas, Muhammad R.: *ARC: A Vision-based Automatic Retail Checkout System.* https://arxiv.org/abs/2104.02832. Version: 2021